Cortical thickness - Insight46 phase 1 and 2

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Document details

Categories of variables	Cortical thickness derived for both insight46 phase 1 and 2
Summary of work undertaken	See below for summary of methods and references All cortical thickness values are in millimetres (mm).
Names of source variables	lh_bankssts_area, rh_bankssts_area, lh_bankssts_thickness, rh_bankssts_thickness
Output variables	lh_adm_corthck_i46p1, rh_adm_corthck_i46p1, lh_adh_corthck_i46p1, rh_adh_corthck_i46p1, mean_adm_corthck_i46p1, mean_adh_corthck_i46p1, lh_adm_corthck_i46p2, rh_adm_corthck_i46p2, lh_adh_corthck_i46p2, rh_adh_corthck_i46p2, mean_adm_corthck_i46p2, mean_adh_corthck_i46p2
Papers using these variables	Keuss SE., et al. Rates of cortical thinning in Alzheimer's disease signature regions: pathological influences and cognitive consequences in members of the 1946 British birth cohort. Abstract submitted for AAIC 2022 and accepted as oral presentation. Paper based on the above abstract currently in progress.

Overview

<u>For Submission to the NSHD Scientific Support Team</u>

Methods

T1 MRI data from each time-point underwent correction for gradient non-linearity (Jovicich et al., 2006), and were manually checked by a trained team who assessed motion, anatomical coverage, and other quality control issues. Scan-pairs were also reviewed to check for unacceptable differences in quality between scans.

Phase 1 T1 MRI data that passed cross-sectional quality control (n=468) and phase 2 T1 MRI data that passed cross-sectional and longitudinal quality control (n=356) were processed using Freesurfer 7.1.0 (). Images from each time-points first underwent cross-sectional processing, including motion correction, removal of the skull, spatial transforms, intensity normalisation, cortical surface reconstruction and parcellation (Dale, Fischl and Sereno, 1999; Fischl, Sereno and Dale, 1999; Fischl and Dale, 2000; Desikan et al., 2006).

Cross-sectionally processed images were then run through the longitudinal stream in Freesurfer (Reuter et al., 2012). Specifically, an unbiased within-subject template was created based on data from both time-points using robust inverse, consistent registration (Reuter, Rosas and Fischl, 2010; Reuter and Fischl, 2011), and then several processing steps were initialised using common information from this template (Reuter et al., 2012).

All cortical thickness segmentations were visually inspected for major errors.

Cortical thickness was assessed in two composite regions of interest that have previously been shown to be vulnerable in early AD: one comprised of middle temporal, inferior temporal, entorhinal and fusiform areas (ADsig Mayo) (Jack et al., 2015); and one consisting of parahippocampal, inferior temporal, entorhinal, temporal pole, precuneus, supramarginal, superior and inferior parietal, superior frontal, parsopercularis, parstriangularis and parsorbitalis areas (ADsig Harvard) (Dickerson et al., 2009).

The regions were formed using Freesurfer's mri_mergelabels command to merge anatomical labels from the Desikan-Killian atlas (Desikan et al., 2006), and the mris_label2annot command to create a single annotation file. Cortical thickness values were then generated using Freesurfer's mris_anatomical_stats command and extracted using the aparcstats2table command. The following formula was used to calculate surface area weighted averages of left and right hemisphere values: (left surface area left thickness + right surface area right thickness) / left + right surface area.

This work was led by Sarah Keuss () and Will Coath () with senior guidance from Dave Cash ().

References

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Reuter, M., Rosas, H. D. and Fischl, B. (2010) ‘Highly accurate inverse consistent registration: A robust approach’, NeuroImage. Neuroimage, 53(4), pp. 1181–1196. doi: 10.1016/j.neuroimage.2010.07.020.

Variables in this document

12 high-confidence matches

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Brain MRI [6033] › Cortical thickness [60334]		
lh_adh_corthck_i46p1	Left hemisphere ADsig Harvard cortical thickness (mm) - Longitudinal - Insight46 phase 1	topsheet field
lh_adh_corthck_i46p2	Left hemisphere ADsig Harvard cortical thickness (mm) - Longitudinal - Insight46 phase 2	topsheet field
lh_adm_corthck_i46p1	Left hemisphere ADsig Mayo cortical thickness (mm) - Longitudinal - Insight46 phase 1	topsheet field
lh_adm_corthck_i46p2	Left hemisphere ADsig Mayo cortical thickness (mm) - Longitudinal - Insight46 phase 2	topsheet field
mean_adh_corthck_i46p1	Surface area weighted average of left and right hemisphere ADsig Harvard cortical thickness (mm) - Longitudinal - Insight46 phase 1	topsheet field

Variable	Label	How it was found
mean_adh_corthck_i46p2	Surface area weighted average of left and right hemisphere ADsig Harvard cortical thickness (mm) - Longitudinal - Insight46 phase 2	topsheet field
mean_adm_corthck_i46p1	Surface area weighted average of left and right hemisphere ADsig Mayo cortical thickness (mm) - Longitudinal - Insight46 phase 1	topsheet field
mean_adm_corthck_i46p2	Surface area weighted average of left and right hemisphere ADsig Mayo cortical thickness (mm) - Longitudinal - Insight46 phase 2	topsheet field
rh_adh_corthck_i46p1	Right hemisphere ADsig Harvard cortical thickness (mm) - Longitudinal - Insight46 phase 1	topsheet field
rh_adh_corthck_i46p2	Right hemisphere ADsig Harvard cortical thickness (mm) - Longitudinal - Insight46 phase 2	topsheet field
rh_adm_corthck_i46p1	Right hemisphere ADsig Mayo cortical thickness (mm) - Longitudinal - Insight46 phase 1	topsheet field
rh_adm_corthck_i46p2	Right hemisphere ADsig Mayo cortical thickness (mm) - Longitudinal - Insight46 phase 2	topsheet field